Lean 4 Verification Report: Spherical N-Series Area Law — Theorem 2 on a Regular Patch

Nov 11, 2026 · @Wayne Baker

Summary

Theorem 2 of the v2.4 draft, restricted to a regular patch, is now a Lean 4 theorem: `BakerSpherical.AreaLaw.spherical_area_law`. Its corollary `richardson_fourth_order` gives the fourth-order Richardson result. Both compile with exit code 0 on the author's Windows machine and depend only on `propext`, `Classical.choice` and `Quot.sound`.

The final theorem rests on 31 Lean files containing 126 audited declarations. No quadrature estimate, Taylor remainder or derivative bound is assumed; each is proved in the chain.

This supersedes the scope of the Nov 10 report, which verified supporting algebra and the smooth extension only. The result does not cover a patch containing a coordinate pole, the sixth- or eighth-order extrapolations, or the Arb-certified numerical constants. The Scope section states the boundary.

The theorem

Hypotheses. F maps the plane to Euclidean 3-space and is of class C^5. On the closed unit parameter square, the cross product $F_s \times F_t$ is nonzero at every point.

Grid. The square is cut into N × N cells of side h = 1/N with midpoint centres $x_{ij} = ((i + 1/2)h, (j + 1/2)h)$. Each cell is split along its lower-left to upper-right diagonal into two chordal triangles. A_N (`chordalArea F N`) is the total area of those triangles.

Statement (`spherical_area_law`). There exist H > 0 and C ≥ 0 such that for every N ≥ 1 with 1/N ≤ H:

```
\left| A_N - \left( A + \frac{C_2}{N^2} \right) \right| \le \frac{C}{N^4}
```

Corollary (`richardson_fourth_order`). With the same H and C:

```
\left| \frac{4A_{2N} - A_N}{3} - A \right| \le \frac{5}{12}\,\frac{C}{N^4}
```

Symbol	Lean name	Definition
J	`jac F`	Norm of F_s × F_t
A	`areaIntegral F`	Iterated integral of J over the unit square
C_2	`areaC2 F`	$-(1/24)(\iint J_{ss} + \iint J_{tt}) + \iint q_2$
q_2	`q2 F y`	$\langle r, N\rangle / (48\|N\|) + \langle F_{st}, N\rangle^2 \, \|F_s + F_t\|^2 / (8\|N\|^3)$
N	`normalField F y`	F_s × F_t
r	(inline)	$(F_s - F_t) \times D_+^3 F + D_-^3 F \times (F_s + F_t)$

$D_+^3 F$ and $D_-^3 F$ are the third derivatives along the two diagonals. `SphericalMixedPartials.lean` proves they equal F_sss ± 3F_sst + 3F_stt ± F_ttt, and that `Fst` equals the mixed partial ðsðt F.

Environment

Item	Value
Lean toolchain	`leanprover/lean4:v4.36.0-rc2`
Mathlib	commit `ccd6c8476363fc81f2e4bee2eeb1a62c99338549`
Project	`C:\Users\Wayne\Nov_26\SphericalNew\SphericalLemma2`
Primary build	Author's Windows machine, full `import Mathlib`, run 2026-11-11
Second build	Linux, same toolchain and Mathlib commit built from source, reduced import lists

Any statement citing this verification should quote the Mathlib commit hash, not a tag. The Windows build is the build of record. The Linux build used specific Mathlib imports in place of `import Mathlib`, so it is a cross-check and not an identical reproduction.

Ledger A: files the final theorem depends on

These 31 files are the import closure of `SphericalAreaLaw.lean`. "Audited" counts the declarations each file prints with `#print axioms`; all 126 show the three standard axioms and no `sorryAx` in the Windows logs.

Stage	File	Audited	Role
Chords	`NormalizedChordSmoothExtension`	36	Chord quotient; pairedArea = $h^2 \cdot$ normArea; evenness; smoothness through h = 0
Cross product	`CrossProductContinuous`	2	Cross product as a continuous bilinear map on coordinates
Cross product	`CrossProductEuclidean`	1	Transport to Euclidean 3-space
Cross product	`CrossProductBilinear`	2	`crossE` and its alternating property
Cross product	`CrossProductAreaBridge_v2`	4	Abstract area results specialized to `crossE`
Regularity	`CrossProductRegularity`	1	Smoothness of normArea away from h = 0
Regularity	`NondegenerateAreaDomain`	1	The regular domain is open
Regularity	`CompactUniformWidth`	1	Uniform width over a compact set
Regularity	`SphericalUniformWidth`	1	Uniform nondegenerate cell width
Regularity	`UniformAreaRegularity`	1	C^4 regularity on a compact tube
Regularity	`CompactUniformBound`	1	Bound for a continuous function on a compact set
Regularity	`SphericalFourthDerivative`	1	Uniform bound on the fourth Fréchet derivative
Slices	`SphericalVerticalDirection`	2	The h-direction unit vector

Stage	File	Audited	Role
Slices	`SphericalVerticalFourthBound`	2	Operator-norm bound along the h-direction
Slices	`SphericalSliceFourthIdentity`	5	Affine-slice chain rule from local regularity
Slices	`SphericalSliceFourthBound`	2	Fourth derivative of each h-slice bounded by M
Taylor	`SphericalTaylorSlice`	2	`sliceArea`, its evenness and value at 0
Taylor	`EvenTaylorCoefficients`	2	Odd derivatives of an even function vanish at 0
Taylor	`TaylorDerivativeBridge`	1	Within-derivatives agree with derivatives
Taylor	`TaylorCubicReduction`	1	Cubic Taylor polynomial with vanishing odd terms
Taylor	`TaylorEvenPolynomial`	1	Even Taylor polynomial of order three
Taylor	`TaylorSecondDerivativeBridge`	1	`iteratedDeriv 2` = `deriv (deriv ·)`
q_2	`Lemma2Algebra`	4	Polynomial identity behind Lemma 2
q_2	`SphericalQ2Coefficient`	13	Second h-derivative of normArea at h = 0
q_2	`SphericalQ2ClosedForm`	3	Closed form of q_2 from `q2_polynomial_identity`
q_2	`SphericalMixedPartials`	8	`Fst`, `DP3`, `DM3` identified with partial derivatives
Local	`SphericalLocalTaylor`	4	Single-cell expansion, remainder $M h^4 / 24$
Global	`SphericalCellSum`	3	Summation over the N × N grid

Stage	File	Audited	Role
Quadrature	`MidpointQuadrature1D`	7	Midpoint rule on $[0, 1]$, orders two and four
Quadrature	`MidpointQuadrature2D`	8	Midpoint rule on the unit square, orders two and four
Global	`SphericalAreaLaw`	5	Smoothness of J and q_2; the area law; Richardson
	Total	**126**	

Release audit (2026-11-11). The missing SphericalQ2Coefficient build log was regenerated. All 31 modules compiled with exit code 0; all 126 printed declaration audits passed with no reported errors or unexpected axioms. The frozen 66-file proof package and SHA-256 manifest were committed byte-for-byte to the v2.4-draft branch. Release commit: f02f10ea9244a7b8ed5bd063e5a57f0b27c5e09a.